

Requirement Specification Document

E-Commerce Order Management Database System

1. Introduction

1.1 Purpose

This document describes the business requirements of an E-Commerce Order Management System and specifies the data, functional, reporting and quality requirements of the database that will support it. It is the basis for the later phases of the project: ER modelling, relational schema design, normalization, SQL implementation and PL/SQL automation.

1.2 Scope

The system will manage the daily operations of an online shopping platform similar to Amazon or Flipkart. The database will store and process information about customers, products, categories, suppliers, inventory, orders, payments, shipments and customer reviews, and will generate sales reports for management.

In scope: customer management, catalog management, inventory tracking, order processing, payment tracking, shipment tracking, reviews and ratings, and sales reporting.

Out of scope: front-end web or mobile application design, payment gateway integration, and logistics partner integration. The system is a database back end only.

1.3 Definitions and Abbreviations

Term	Meaning
DBMS	Database Management System
ER Diagram	Entity Relationship Diagram
PK / FK	Primary Key / Foreign Key
SQL	Structured Query Language
PL/SQL	Procedural Language extension to SQL
SKU	Stock Keeping Unit (a unique product identifier)
CRUD	Create, Read, Update, Delete

1.4 Intended Audience

Course faculty and the subject matter expert (evaluation), the student database designer and developer (implementation), and, in the business scenario, the e-commerce management team who use the reports.

2. Business Overview

2.1 Problem Statement

A rapidly growing e-commerce company handles thousands of customers and products. Managing this data with spreadsheets or separate files causes duplicate records, inconsistent stock figures, lost orders, mismatched payments and slow reporting. The company needs a centralized, reliable relational database that keeps data accurate, enforces business rules and supports decision-making.

2.2 Business Objectives

1. Maintain a single, consistent store of customer, product and order data.
2. Track stock levels in real time so that out-of-stock items are never sold.
3. Record every order from placement to delivery, with its payment and shipment status.
4. Reduce data redundancy and protect data integrity.
5. Provide accurate sales reports and analytics for management.
6. Automate routine operations such as stock updates and payment checks.

2.3 Stakeholders

Stakeholder	Interest in the system
Customer	Registers, browses products, places and tracks orders, writes reviews
Administrator	Manages customers, categories, products and system data
Inventory / Warehouse Manager	Monitors stock and suppliers, handles replenishment
Sales / Business Manager	Uses sales reports and analytics
Payment / Accounts Team	Verifies payments and handles refunds
Logistics / Shipping Team	Updates shipment and delivery status
Database Designer / Developer	Designs, builds and tests the database (the student)

3. Overall Description

3.1 System Perspective

The database sits at the center of the e-commerce platform. Application screens and reports read from and write to it. It follows the relational model and is implemented using SQL and PL/SQL.

3.2 Typical Business Workflow

1. A customer registers and a profile is created.
2. The customer browses products grouped by category.
3. The customer places an order containing one or more products.
4. The system checks stock, records the order and order items, and calculates the total.
5. The customer pays and the payment is recorded with its status.
6. Stock is reduced after the order is confirmed.
7. The order is shipped and the shipment status is tracked until delivery.
8. After delivery, the customer can rate and review the product.
9. Management generates periodic sales reports.

3.3 Assumptions and Dependencies

- Each order belongs to exactly one customer.
- A product belongs to one category and is provided by one supplier.
- Payment is recorded per order, and one shipment is created per order.
- Only registered customers can place orders and write reviews.
- A customer can review only products that they have purchased (enforced at application or procedure level).
- Prices are stored in a single currency (INR).

3.4 Constraints

- The database must be a relational database implemented with standard SQL and PL/SQL (Oracle or an equivalent DBMS).
- The design must be normalized up to 3NF/BCNF.
- The project is built progressively over a 16-week semester.

4. Functional Requirements

4.1 Customer Management

ID	Requirement
FR-01	The system shall allow new customers to register with name, email, mobile number and address.
FR-02	The system shall automatically record the registration date.
FR-03	The system shall ensure that email and mobile number are unique for every customer.
FR-04	The system shall allow customer profile details to be viewed and updated.
FR-05	The system shall retain a customer's order history even if the profile is later made inactive.

4.2 Category and Product Management

ID	Requirement
FR-06	The system shall store product categories with a name and description.
FR-07	The system shall allow products to be added, updated and removed, each assigned to one category.
FR-08	The system shall store product name, price and stock quantity.
FR-09	The system shall not allow a negative price or negative stock quantity.
FR-10	The system shall allow product prices to be updated while preserving the unit price of past orders.

4.3 Supplier Management

ID	Requirement
FR-11	The system shall store supplier name and contact information.
FR-12	The system shall link each product to the supplier who provides it.
FR-13	The system shall allow the list of products supplied by a given supplier to be retrieved.

4.4 Inventory Management

ID	Requirement
FR-14	The system shall maintain the available stock quantity for every product.
FR-15	The system shall check stock availability before accepting an order item.
FR-16	The system shall reduce stock when an order is placed and restore it when an order is cancelled.
FR-17	The system shall identify low-stock and out-of-stock products.

4.5 Order Management

ID	Requirement
FR-18	The system shall allow a customer to place an order containing one or more products.
FR-19	The system shall record the order date, customer, status and total amount.
FR-20	The system shall store each ordered product with its quantity and unit price in the order details.
FR-21	The system shall calculate the total amount as the sum of (quantity × unit price) of all items.
FR-22	The system shall maintain order status values such as Placed, Confirmed, Shipped, Delivered and Cancelled.

ID	Requirement
FR-23	The system shall allow an order to be cancelled only before it is shipped.
FR-24	The system shall allow a customer's complete order history to be retrieved.

4.6 Payment Processing

ID	Requirement
FR-25	The system shall record the payment method (for example UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery) for each order.
FR-26	The system shall record the payment date and payment status (Pending, Completed, Failed, Refunded).
FR-27	The system shall validate that the payment is linked to a valid order.
FR-28	The system shall mark a payment as Refunded when a paid order is cancelled.

4.7 Shipment Tracking

ID	Requirement
FR-29	The system shall create a shipment record for every confirmed order.
FR-30	The system shall store the delivery address, shipment date and delivery status.
FR-31	The system shall maintain delivery status values such as Processing, Shipped, Out for Delivery, Delivered and Returned.
FR-32	The system shall allow customers and staff to track the status of any shipment.

4.8 Reviews and Ratings

ID	Requirement
FR-33	The system shall allow a customer to rate a product on a scale of 1 to 5 and add comments.
FR-34	The system shall allow a customer to review a product only once.
FR-35	The system shall calculate the average rating and number of reviews for each product.

4.9 Reporting and Analytics

ID	Report	Purpose
FR-36	Monthly Sales Report	Total orders and revenue for each month
FR-37	Product-wise Revenue	Revenue generated by each product
FR-38	Category-wise Sales	Sales distribution across categories
FR-39	Top-Selling Products	Products with the highest quantity sold
FR-40	Highest-Spending Customers	Customers ranked by total purchase amount
FR-41	Customer Purchase History	All orders and items for a customer
FR-42	Low-Stock Report	Products below the reorder level

4.10 Automation and Auditing

ID	Requirement
FR-43	Stock shall be updated automatically when order items are inserted or cancelled (trigger).
FR-44	Payments shall be validated automatically against the order amount and status (trigger).
FR-45	Every change to order status or amount shall be recorded in an audit table (trigger).
FR-46	The system shall provide stored procedures to place an order, cancel an order and update a product price.
FR-47	The system shall provide functions to compute total sales, a customer's purchase amount and inventory availability.
FR-48	The system shall handle errors such as invalid product ID, out-of-stock items and payment failures with meaningful messages.

5. Data Requirements

5.1 Entities and Attributes

Customer

Attribute	Description	Key / Rule
Customer ID	Unique customer identifier	Primary Key
Name	Full name	Not null
Email	Email address	Unique, not null
Mobile Number	10-digit contact number	Unique, not null
Address	Residential address	Not null
Registration Date	Date of registration	Default: current date

Category

Attribute	Description	Key / Rule
Category ID	Unique category identifier	Primary Key
Category Name	Name of the category	Unique, not null
Description	Short description	Optional

Product

Attribute	Description	Key / Rule
Product ID	Unique product identifier	Primary Key
Product Name	Name of the product	Not null
Price	Selling price	Check: price > 0
Stock Quantity	Units available	Check: quantity >= 0
Category ID	Category of the product	Foreign Key to Category
Supplier ID	Supplier of the product	Foreign Key to Supplier

Supplier

Attribute	Description	Key / Rule
Supplier ID	Unique supplier identifier	Primary Key
Supplier Name	Name of the supplier	Not null
Contact Information	Phone / email / address	Not null

Order

Attribute	Description	Key / Rule
Order ID	Unique order identifier	Primary Key
Customer ID	Customer who placed the order	Foreign Key to Customer
Order Date	Date of placing the order	Default: current date
Total Amount	Order value	Check: amount >= 0
Order Status	Current state of the order	Check: allowed status list

Order Details

Attribute	Description	Key / Rule
Order Detail ID	Unique line item identifier	Primary Key

Attribute	Description	Key / Rule
Order ID	Parent order	Foreign Key to Order
Product ID	Ordered product	Foreign Key to Product
Quantity	Units ordered	Check: quantity > 0
Unit Price	Price at the time of purchase	Check: price > 0

Payment

Attribute	Description	Key / Rule
Payment ID	Unique payment identifier	Primary Key
Order ID	Order being paid for	Foreign Key to Order
Payment Method	Mode of payment	Check: allowed methods
Payment Date	Date of payment	Default: current date
Payment Status	Result of payment	Check: allowed status list

Shipment

Attribute	Description	Key / Rule
Shipment ID	Unique shipment identifier	Primary Key
Order ID	Order being shipped	Foreign Key to Order
Delivery Address	Address of delivery	Not null
Shipment Date	Date of dispatch	Optional until shipped
Delivery Status	Current delivery state	Check: allowed status list

Review

Attribute	Description	Key / Rule
Review ID	Unique review identifier	Primary Key
Customer ID	Reviewer	Foreign Key to Customer
Product ID	Product reviewed	Foreign Key to Product
Rating	Score from 1 to 5	Check: rating between 1 and 5
Comments	Review text	Optional

5.2 Relationships Between Entities

Relationship	Type	Explanation
Customer places Order	One-to-Many	One customer can place many orders; each order belongs to one customer
Category contains Product	One-to-Many	One category has many products; each product is in one category
Supplier supplies Product	One-to-Many	One supplier supplies many products
Order contains Product	Many-to-Many	Resolved through Order Details (an order has many products; a product appears in many orders)
Order has Payment	One-to-One (or One-to-Many if partial or retry payments are allowed)	Each order is paid for by a payment record
Order has Shipment	One-to-One	Each order has one shipment record
Customer reviews Product	Many-to-Many	Resolved through Review (a customer reviews many products; a product has many reviews)

5.3 Expected Data Volume

Entity	Minimum records for the project
Customer	50
Category	10
Product	100
Order	100
Supplier, Payment, Shipment, Review, Order Details	Sufficient to support all orders and reports

6. Business Rules

1. Every order must belong to a registered customer.

2. An order must contain at least one product.
3. The quantity ordered cannot exceed the available stock.
4. The unit price in order details is the product price at the time of purchase and does not change if the product price changes later.
5. Order total = sum of (quantity × unit price) for all items in the order.
6. A payment can be made only against an existing order, and the payment amount must match the order total.
7. A shipment can be created only for a paid or confirmed order (Cash on Delivery orders are confirmed on placement).
8. A shipped or delivered order cannot be cancelled.
9. Cancelling an order restores the stock and marks any completed payment as Refunded.
10. A rating must be a whole number between 1 and 5.
11. A customer can submit only one review per product.
12. A category or supplier that has products cannot be deleted.
13. Email addresses and mobile numbers must be unique.

7. Transaction Requirements

Transaction	Steps (summary)
Customer Registration	Validate unique email and mobile, insert customer, set registration date
Product Purchase	Check stock, insert order and order details, compute total, reduce stock
Payment Processing	Validate order, record payment, update order status on success
Order Cancellation	Check status, update order to Cancelled, restore stock, refund payment

Each transaction must be atomic (all steps succeed or none do), keep the data consistent, and be recoverable after failure.

8. Non-Functional Requirements

Category	Requirement
Data Integrity	Primary, foreign, unique, check and default constraints shall be enforced.
Consistency	Stock, order totals and payment statuses shall always agree.
Normalization	The final design shall be in 3NF/BCNF with minimal redundancy.

Category	Requirement
Performance	Common queries (order history, product search, sales reports) shall return quickly; indexes shall be created on key search columns.
Security	Only authorized roles can modify data; customers can view only their own information; sensitive changes are audited.
Reliability	Transactions shall follow ACID properties; backups shall be possible.
Scalability	The design shall support growth in customers, products and orders without restructuring.
Maintainability	Tables, columns and constraints shall use clear, consistent naming.
Usability of Data	Reports shall be easy to generate through SQL views or stored queries.

9. Module-wise Mapping of Project Work

Phase	Weeks	Output
Requirement analysis, ER modelling, relational schema	1 to 3	Requirement report, ER diagram, schema
Functional dependencies and normalization	4 to 6	FD document, normalization report, final design
NoSQL study and comparison	7 to 8	SQL vs NoSQL report, JSON catalog
Database creation, constraints, data, queries	9 to 13	SQL script, query output, analytics report
PL/SQL procedures, functions, cursors, triggers	14 to 15	PL/SQL programs, testing report
Final testing and demonstration	16	Documentation and presentation

10. Conclusion

This document analyzes the business needs of an e-commerce company and translates them into clear functional, data, business-rule and quality requirements. It identifies the nine core entities, their attributes, keys and relationships, and defines the reports and automation the database must support. It will be used as the foundation for designing the ER diagram, deriving the relational schema, normalizing the design and finally implementing the complete E-Commerce Order Management Database System in the coming weeks.